



Commissioning a High-Repetition Rate, 1 Joule Nd:YAG Laser for the HSX Thomson Scattering System

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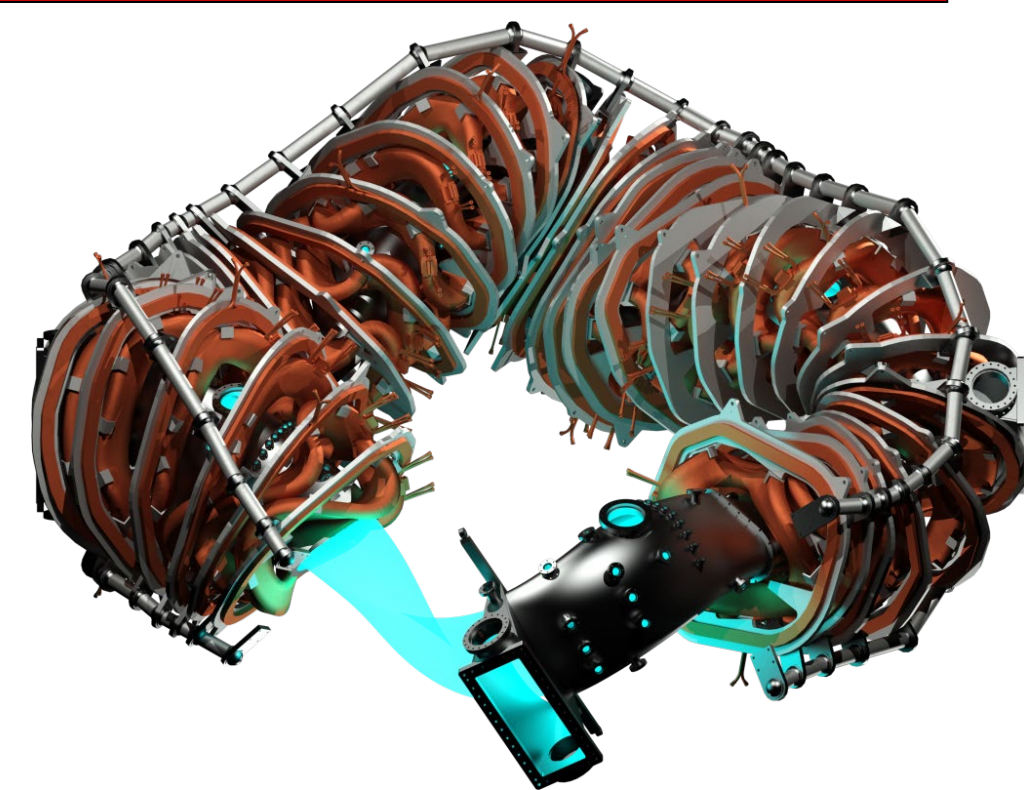
Motivation

We report on the installation and commissioning of a major TS upgrade at the HSX stellarator, centered on a new Nd:YAG laser capable of burst operation of 80 consecutive 1 J, 10 ns pulses at 400 Hz. A 2 kHz mode supports up to 10 pulses per burst, limited by thermal loading. The upgrade includes integrating the laser with HSX beam-delivery optics and shot-to-shot pulse-energy monitoring to verify stability. To accommodate the increased data rate, a high-speed digitizer (3.2 GSa/s) and low-noise preamplifiers, providing an ~8.6 dB improvement in signal-to-noise ratio over the legacy digitizer chain [1,2]. In the near term, increasing the time resolution of the TS system will enable the study of plasma breakdown effects, MHD effects, impurity injection behavior, and time-resolved density profile adjustments.

Thomson Scattering on HSX

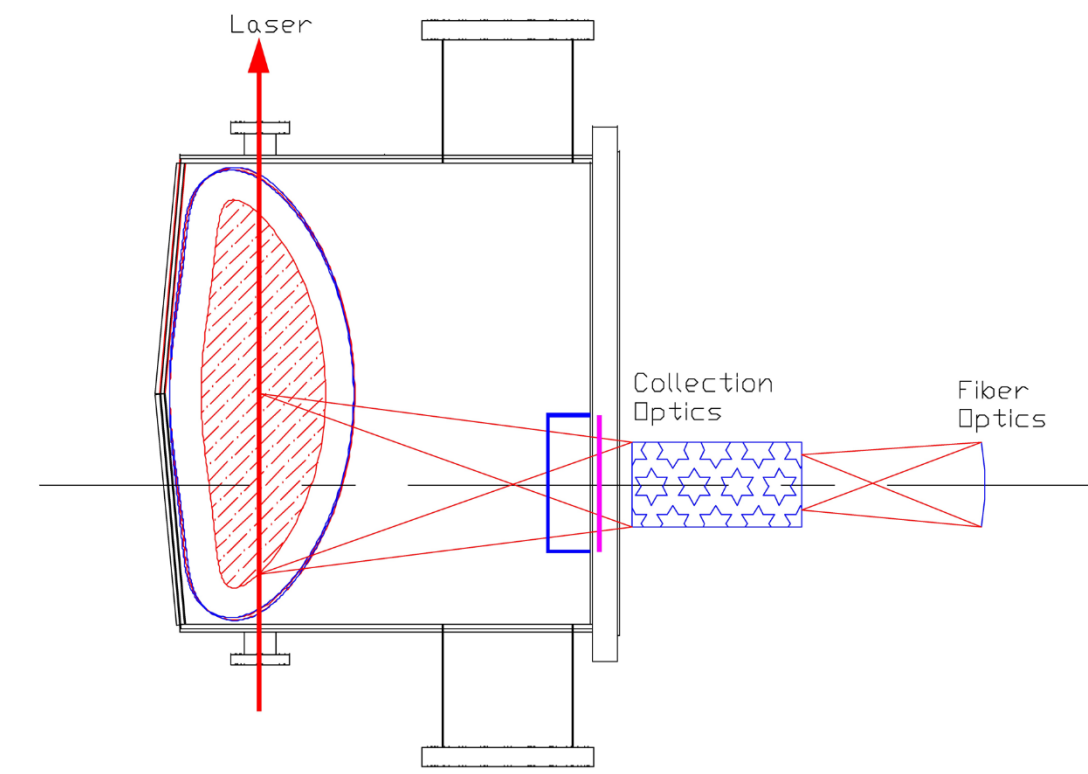
The HSX Stellarator

Major radius: 1.20 m
Average minor radius: 0.12 m
Plasma volume: 0.44 m³
Field periods: 4
Number of main coils: 48
Number of auxiliary coils: 48



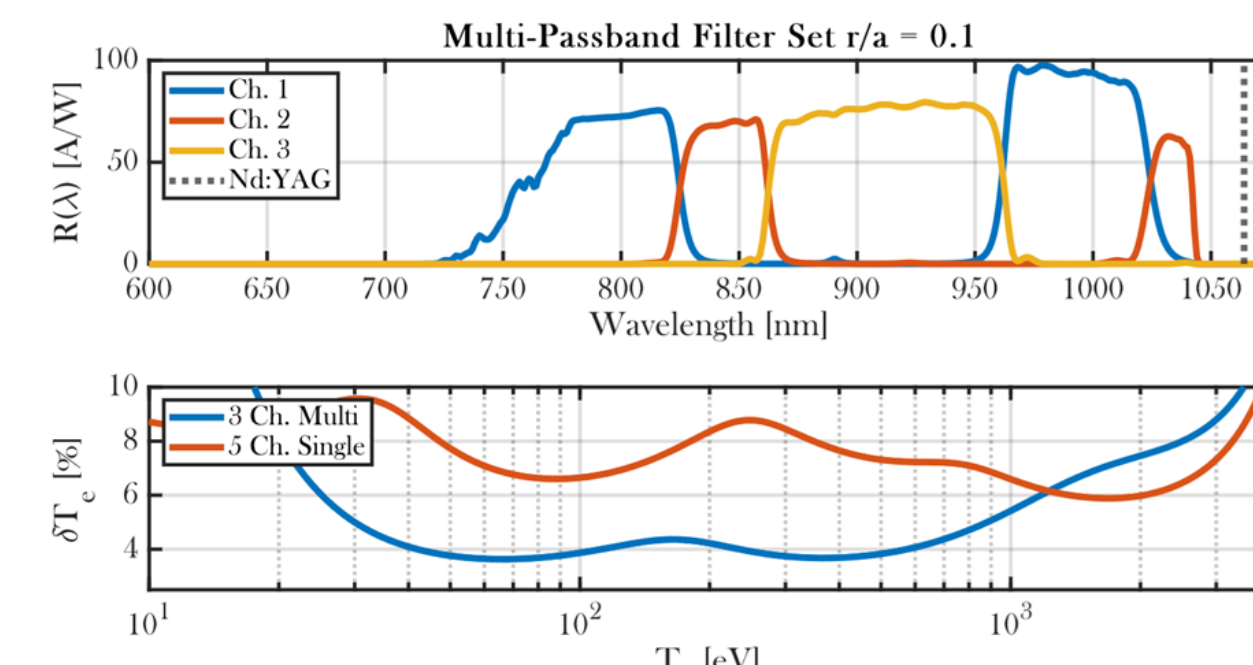
HSX TS System

10 spatial channels [1]
1.6*20 mm scattering area
0.01 ≤ T_e ≤ 4 keV
0.5 ≤ n_e ≤ 10 [10¹⁸ m⁻³]
78.4 MHz detection BW
3.2 GSa/s digitization



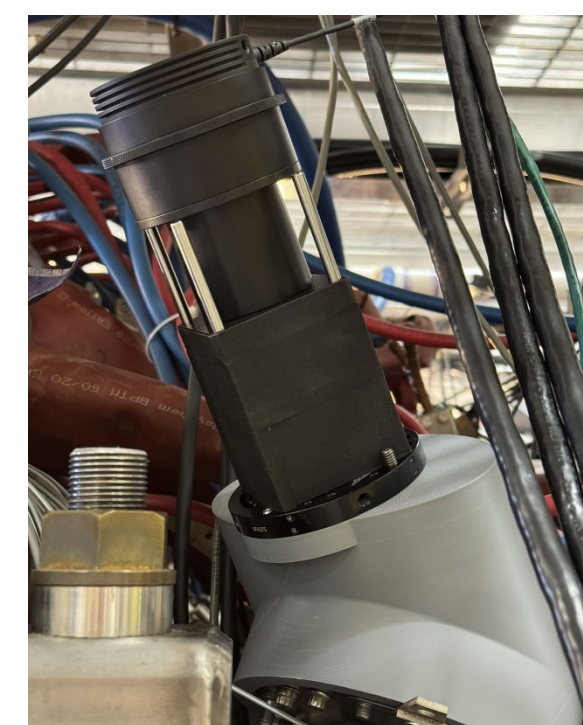
TSSM Filters[3]

Machine learning optimized dual-passband filter set. Reduces error and/or increases T_e bandwidth by more efficiently using the collected photons from a Thomson scattering event.



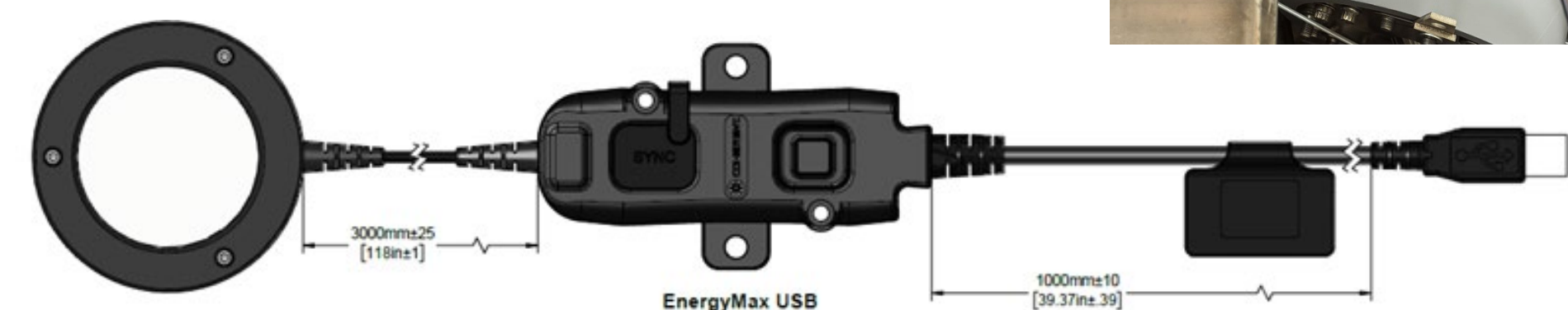
Beam Path

Even with high-reflectivity mirrors, each interaction results in a fractional power loss. Laser was mounted in a manner that there is only **one** optic interaction along the beam path.

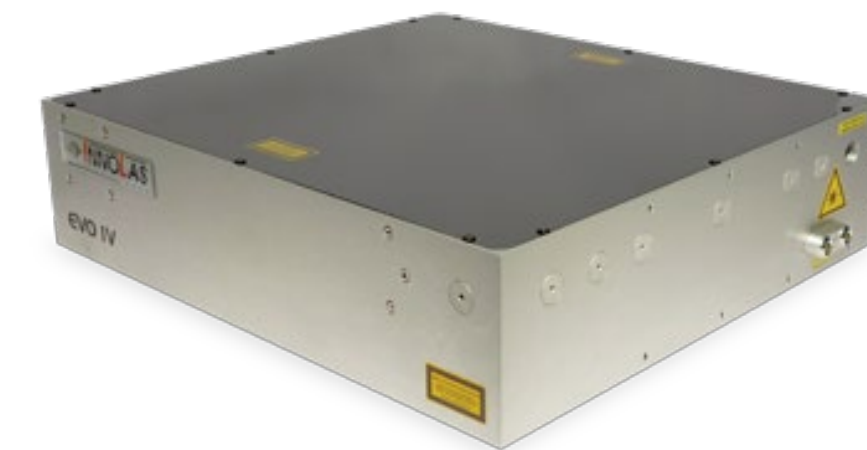


Beam Dump

A Coherent J-50MB-YAG-1528 Energy Monitor is used as a combination beam dump and shot-to-shot energy measurement system enabling individual energy measurements for each TS laser pulse.



Laser Implementation & Results

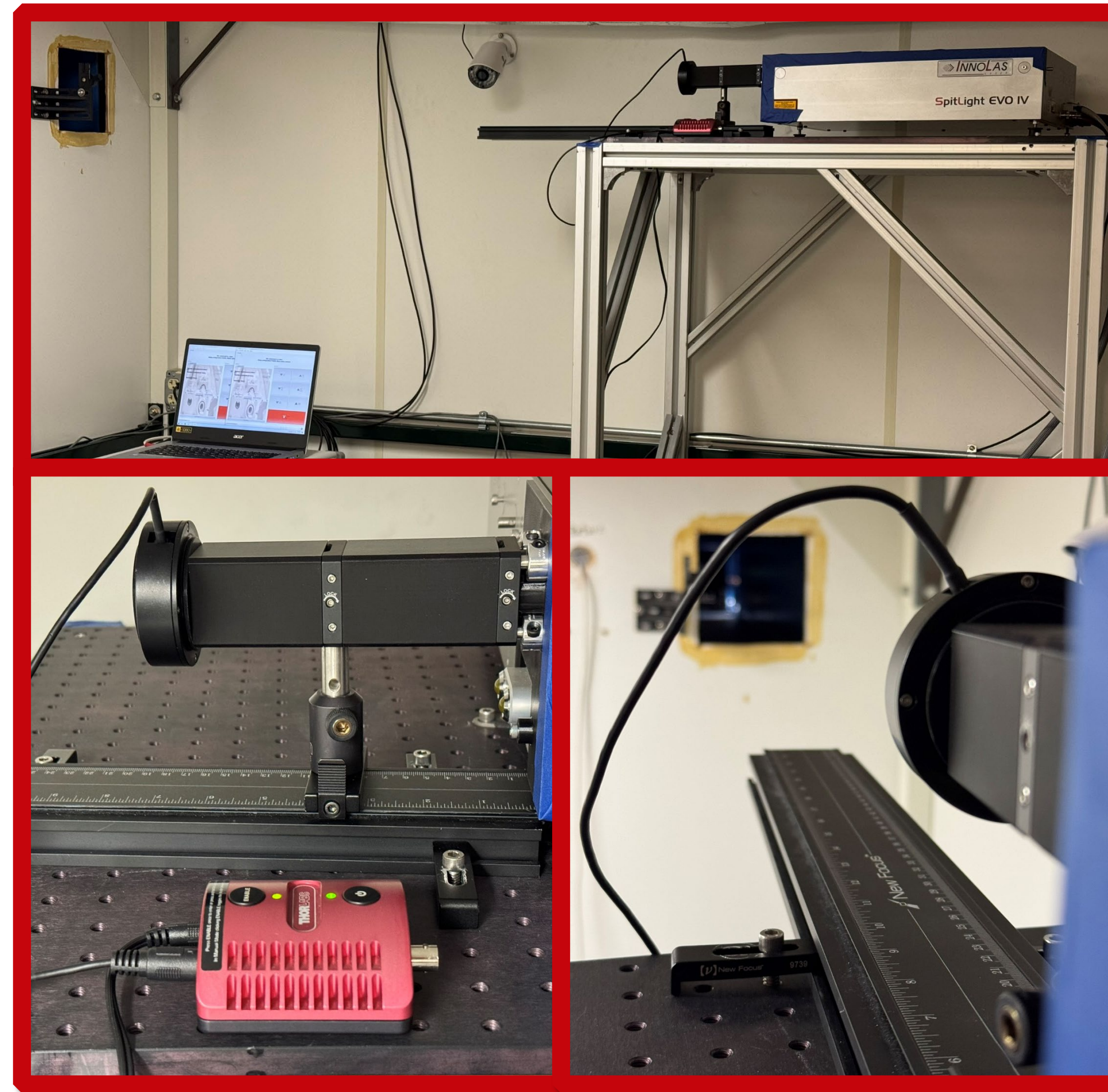


InnoLas Laser System

SpitLight DPSS EVO IV Nd:YAG
~1 Joule, 1064 nm, 8 ns pulse
100 Hz continuous operation
80 pulses @ 400 Hz, 4 x 10 pulses @ 2 kHz
Cold-rod operation standard

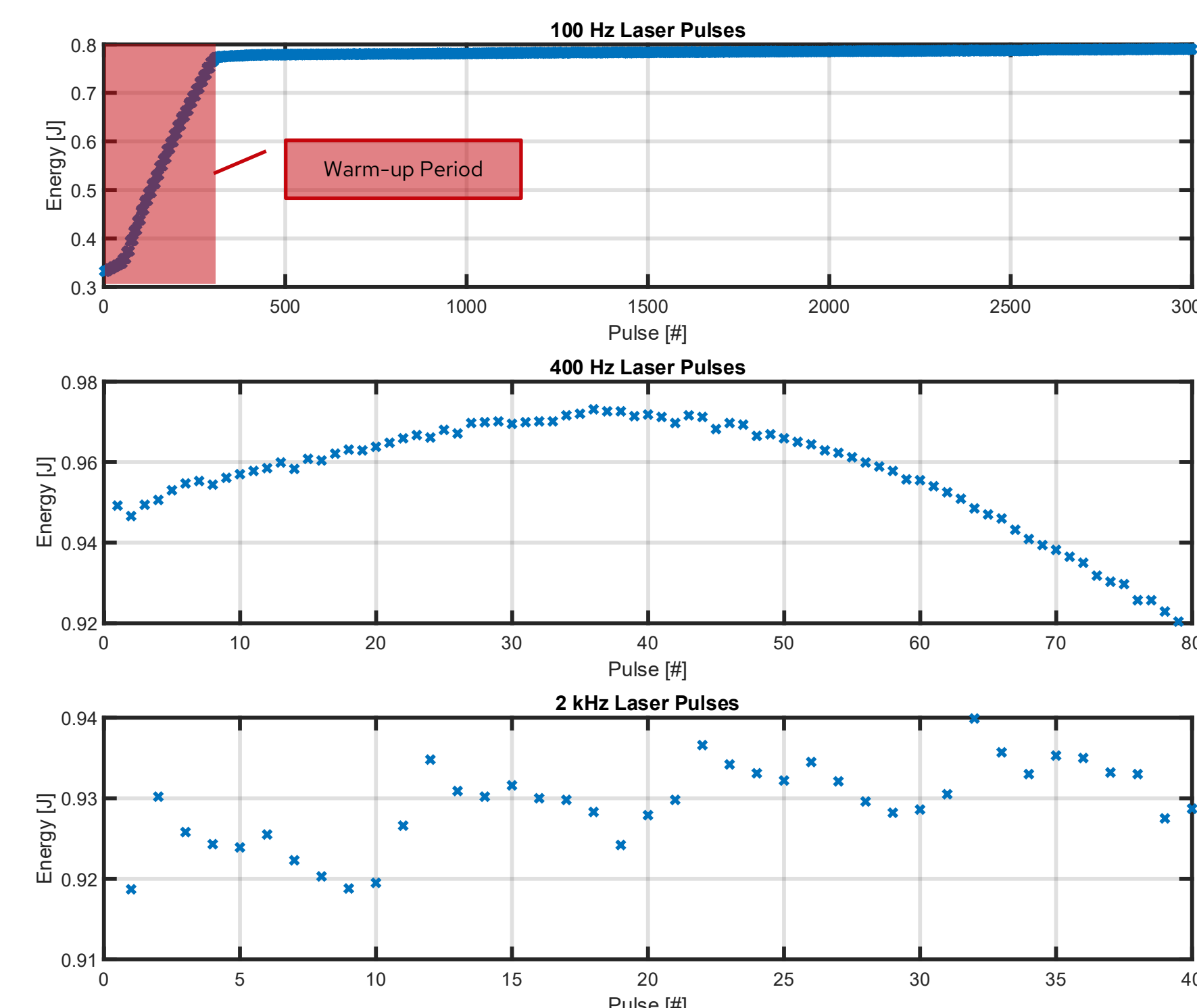
Laser Mounting

The EVO IV was mounted on an optical table ~1 m above the main TS optical table, eliminating at least two optical elements required for a beam periscope. Shown are a $\lambda/4$ -waveplate (QWP), a $\lambda/2$ -waveplate (HWP), and an interlocking beam shutter on the EVO IV exit.



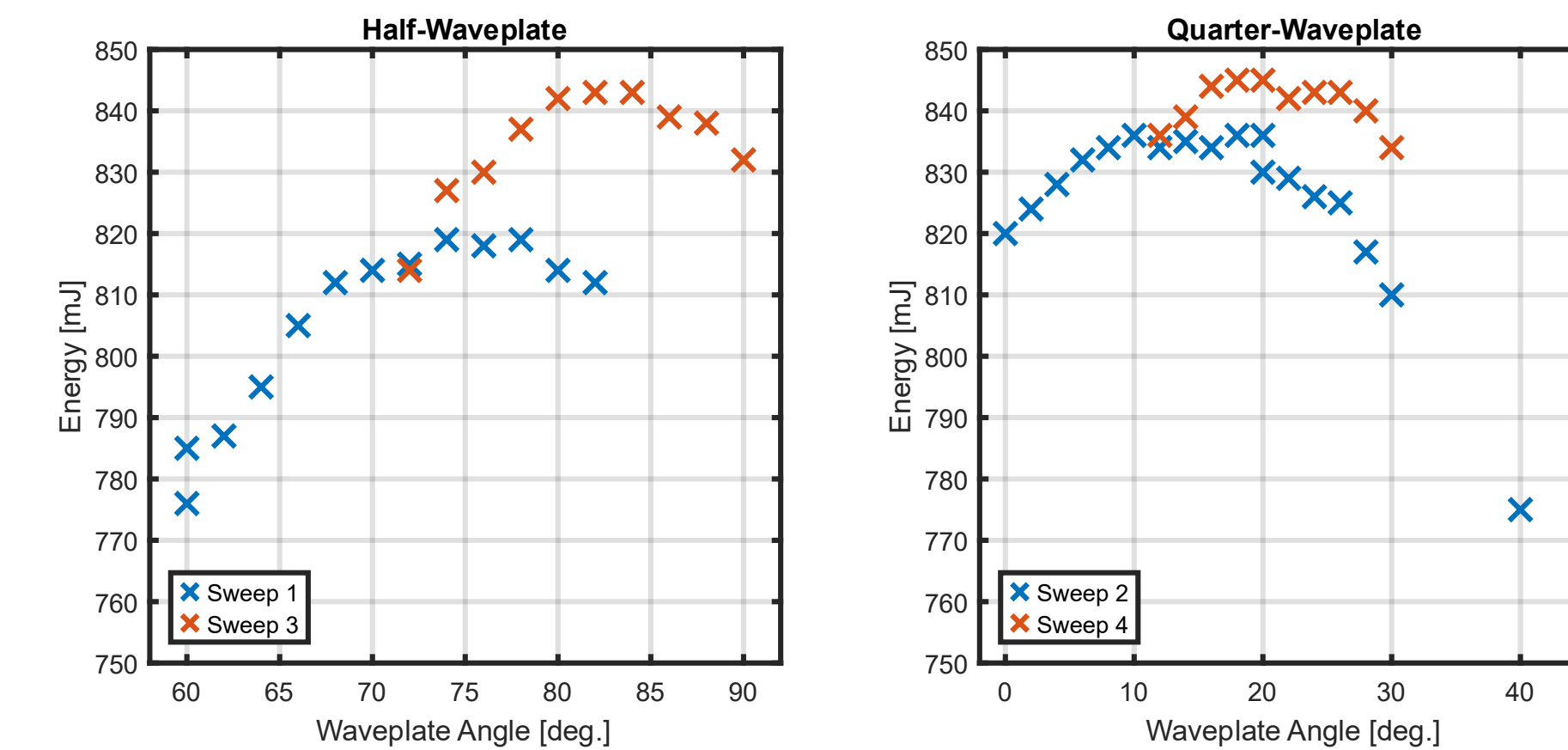
Beam Energy

Beam energies were measured with a Coherent J-50MB-YAG-1528 energy sensor (100 Hz / 400 Hz) or a Coherent J-50MT-10KHZ energy sensor (2 kHz). The SpitLight IV is capable of continuous operation of 100 Hz. 100 Hz operation has a short warm-up period of ~300 pulses before leveling off at maximum energy output. Burst mode operation is enabled for up to 80 pulses at 400 Hz or four sets of ten pulses at 2 kHz. Burst mode operation does not have the initial warm-up period and does not have the same thermal lensing losses seen in 100 Hz operation.



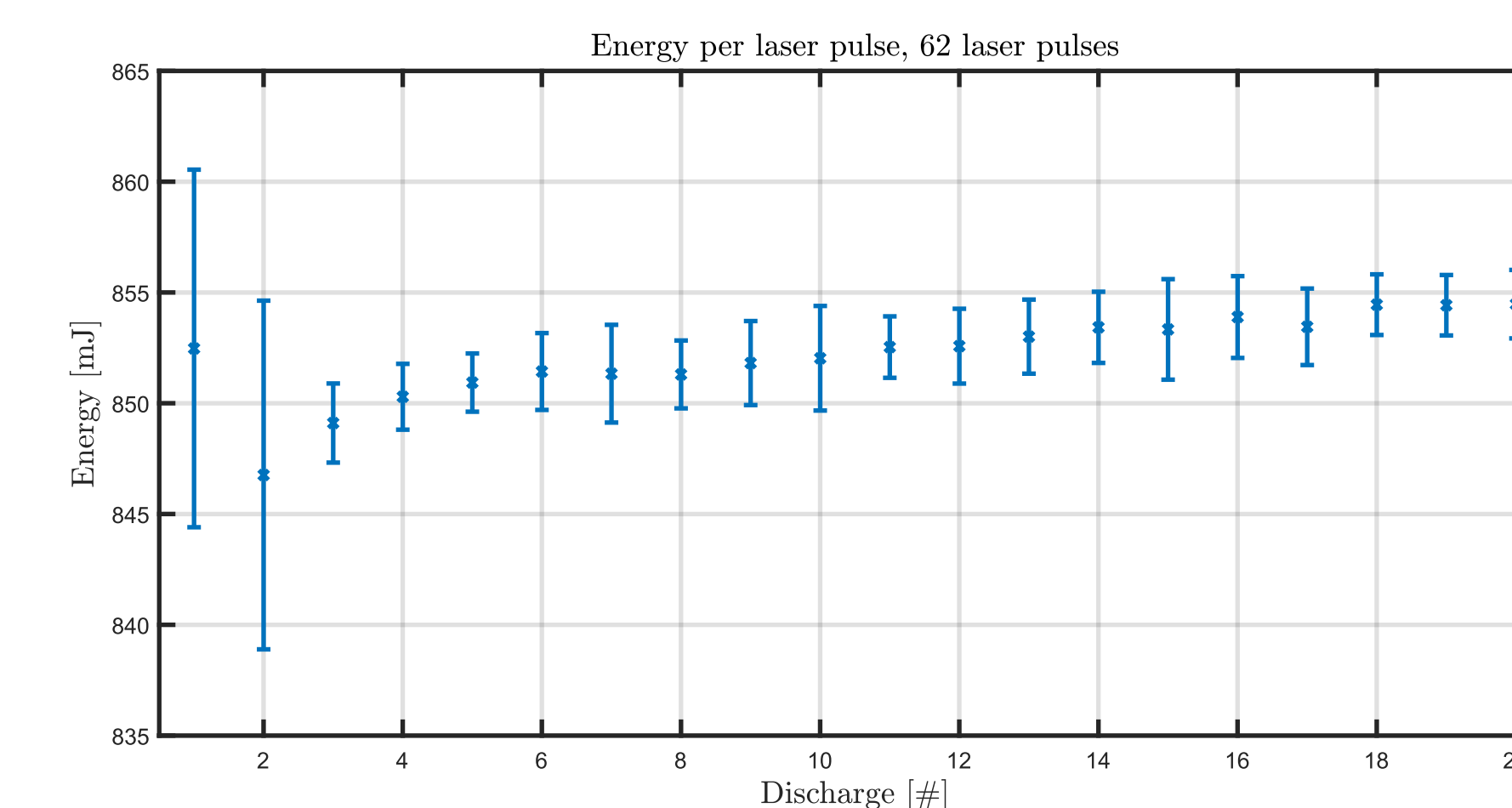
Beam Polarization

Referring to the image of the HSX TS beam dump, a ~6" Brewster window is used as the beam exit. The Brewster window is installed so that p-polarized light is aligned with the magnetic axis, $\vec{B}_{ax}$, which aids in polarization alignment. The one mirror that the HSX TS beam interacts with is at a non-45-degree angle, introducing ellipticity into the beam polarization. Polarization is corrected by adjusting a QWP and HWP in the beam path with the goal of maximizing measured energy through the Brewster window.



Final settings: $\lambda/2$ waveplate @ 82° and $\lambda/4$ waveplate @ 18°.
Uncorrected beam $\theta = 82^\circ$ and phase shift $\delta = 7\pi/64$

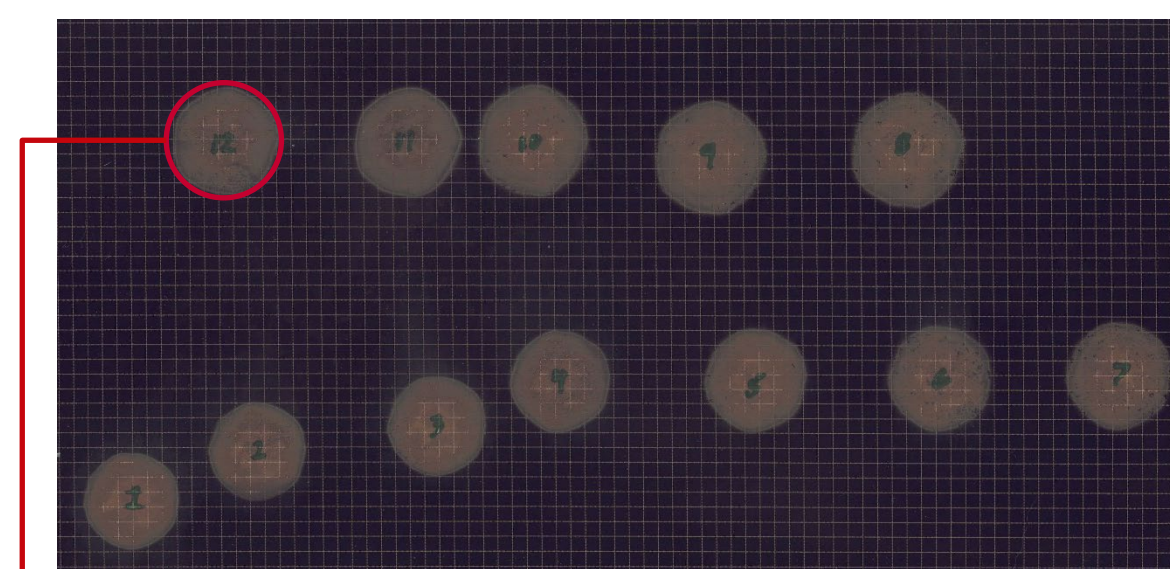
Energy Through Vessel



Warm-rod vs. Cold-rod Operation

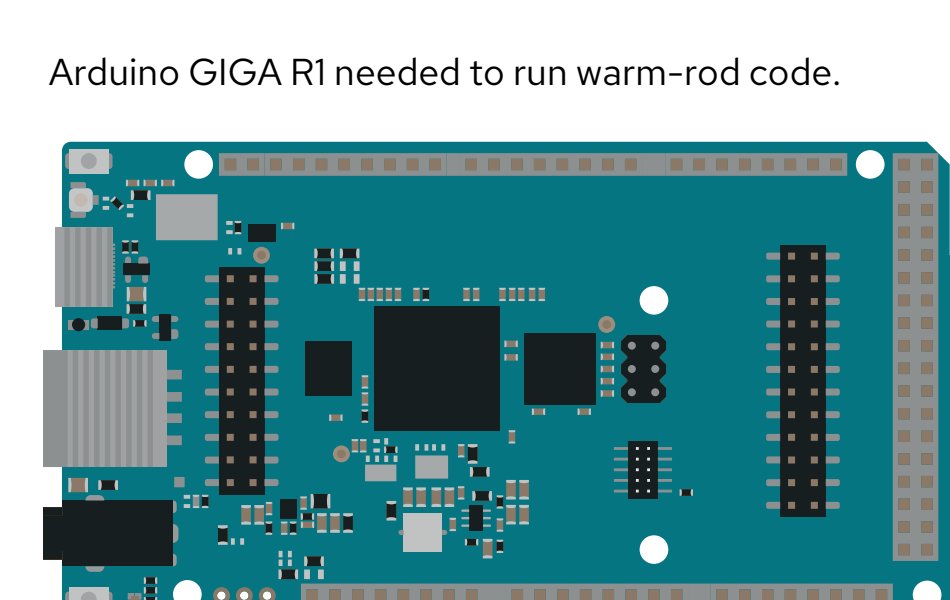
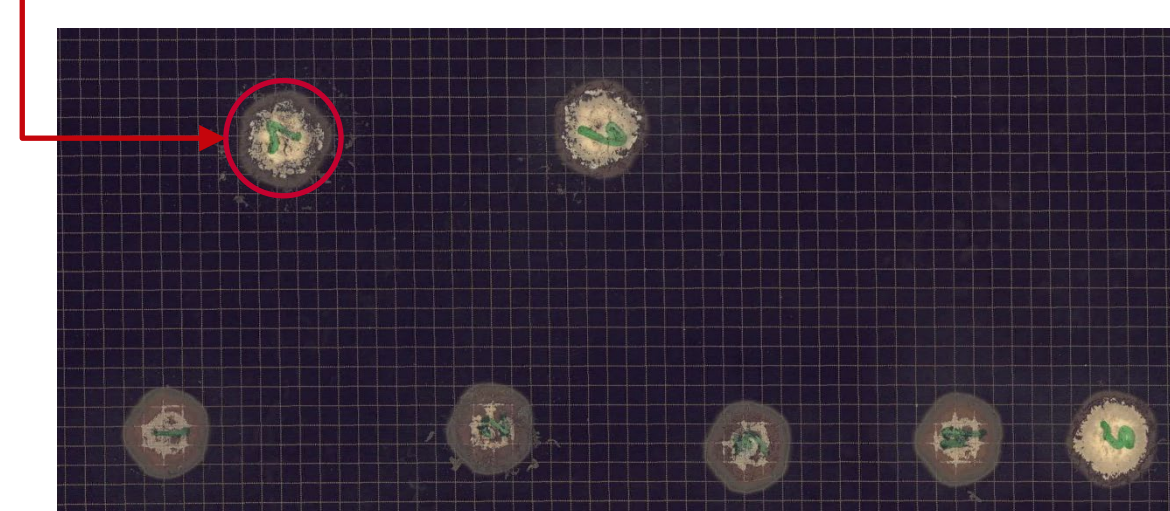
When a diode-pumped solid-state laser operates with cold rods the thermal lens effect within the gain medium is minimal, meaning the rod behaves closer to a flat optical element, which can result in a larger or poorly matched mode volume. As the rods warm under sustained pumping, a roughly parabolic radial temperature gradient develops due to waste heat, creating a positive thermal lens that effectively shortens the cavity's equivalent focal length. A well-designed resonator is typically optimized for this steady-state thermal lens, so warm-rod operation yields a stable, well-defined mode, predictable beam divergence, and consistent output power. For example, for 100 Hz continuous operation, the InnoLas laser system incorporates a warm-up period before the laser functions at the factory design point.

Cold-rod measurements



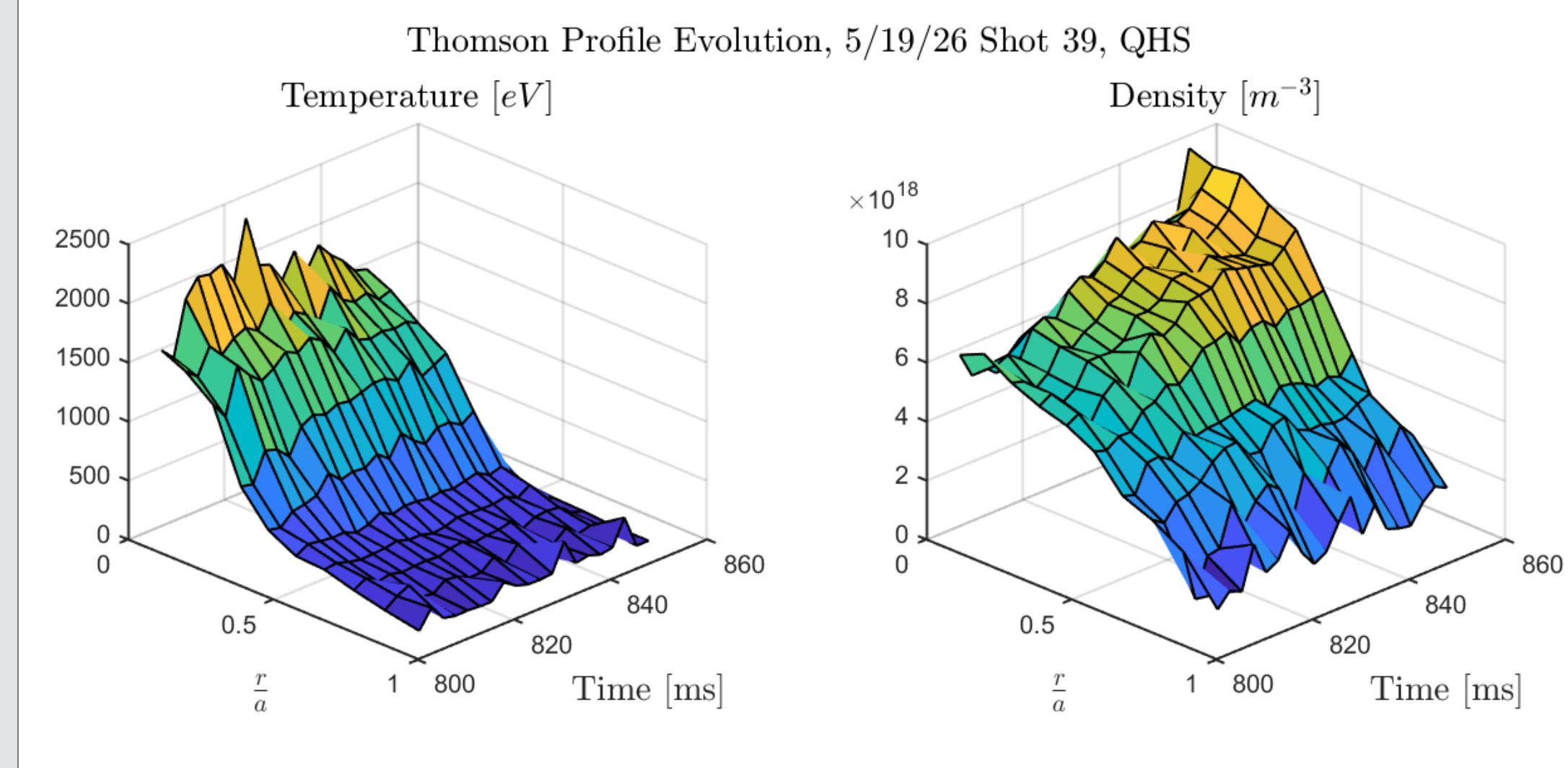
Distance from beam exit [cm]	Beam Dia., ϕ [mm]	
	Cold-rod	Warm-rod
40	14.8	9.76
50	15.24	9.80
60	16.04	10.12
70	16.34	10.16
80	17.15	10.30
90	16.65	10.62
100	17.22	10.72

Warm-rod measurements



Results

An InnoLas SpitLight DPSS EVO IV has been incorporated into daily operations on the HSX Thomson scattering system. Mean beam energy through vessel = 852 [mJ], or 89.7% of initial 950 [mJ] laser output.



Future Work

Commission 2 kHz burst mode for daily operation

Design an ultra-low reflection beam expander lens system to eliminate possible stray light and avoid damage to the Coherent EnergyMax-USB J-50MT-10KHZ Energy Sensor

Commission warm-rod mode for daily operation

Add control channels to the HSX control system, verify operational stability, reperform Raman alignment and calibration.

Info & Resources

For access to all data, materials, and MATLAB functions used to generate this presentation, including code to enable warm-rod operation on an InnoLas laser, please scan the QR code or go to <https://github.com/goodystacks/htpd2026>[4].

For more info on HSX, please visit <https://hsx.wisc.edu/>.

For more info on the University of Wisconsin or the Nuclear Engineering & Engineering Physics department, visit engineering.wisc.edu/departments/nuclear-engineering-engineering-physics/



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References

- [1] K. Zhai et al, Rev. Sci. Instrum., **75**, 3900-3902 (2004)
- [2] W.R. Goodman et al, Rev. Sci. Instrum., **94**, 093507 (2024)
- [3] W. R. Goodman, B. Geiger, Rev. Sci. Instrum., **95**, 093507 (2024)
- [4] W. R. Goodman, GitHub Repo, doi.org/10.5281/zenodo.20269598 (2026)

Acknowledgements

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